

GA-FT User Manual

GA-FT is a command-line tool for fairness testing of binary classifiers. It produces **Individual Discriminatory Instances (IDIs)** — input pairs that differ only in a sensitive attribute but are classified differently by the model under test.

1. Install

```
pip install -r requirements.txt
```

2. Train sklearn models (one-off)

```
python -m ga_fairness_tester.train_sklearn_models
```

This trains Logistic Regression, Random Forest and Gradient Boosting on each dataset and saves them to `models/`. The pre-trained DNNs in `DNN/` are used directly — they require no training step.

3. Run a single fairness test

```
python -m ga_fairness_tester.main \
  --dataset adult \
  --model dnn \
  --sensitive gender \
  --algorithm both \
  --budget 1000 \
  --seed 0
```

Flags

Flag	Choices	Description
--dataset	adult, compas	Dataset under test
--model	dnn, lr, rf, gbm	Model family under test
--sensitive	dataset-dependent (e.g. gender)	Sensitive attribute to protect
--algorithm	rs, ga, both	RS = baseline; GA = GA-FT
--budget	int	Max evaluations per run
--seed	int	Random seed

Sensitive features per dataset: - **adult**: gender, race, age - **compas**: sex, race

4. Output

Each run prints the IDI ratio, IDI count, IDI diversity, and runtime:

```
RS      | IDI_ratio=0.0700 | n_idis= 35 | diversity=0.4123 | 0.1s
GA-FT   | IDI_ratio=0.1500 | n_idis= 75 | diversity=0.5021 | 0.2s
```

5. Running the tests

```
pytest tests/ -q
```

24 unit tests cover the oracle (IDI detection, deduplication, multi-valued sensitive attributes), fitness function (normalisation, distance bonus, α/β arithmetic), GA operators (tournament selection, crossover preserves sensitive feature, mutation respects domains), and statistics (Vargha-Delaney A12 edge cases, magnitude labelling).

6. α/β sensitivity ablation

```
python -m ga_fairness_tester.ablation \
  --dataset adult --model gbm --sensitive gender \
  --seeds 10 --budget 1000
```

Sweeps $\alpha \in \{0.0, 0.3, 0.5, 0.7, 0.9, 1.0\}$ with $\beta = 1 - \alpha$, writing `results/ablation.csv` and `figures/ablation_alpha.png`.

7. Adding a new dataset

1. Drop the pre-processed CSV into dataset/.
2. (Optional) Drop a Keras .h5 model into DNN/.
3. Add an entry to DATASET_CONFIG in [ga_fairness_tester/dataset_config.py](#) with the sensitive feature column indices.
4. Re-run `train_sklearn_models` to train sklearn families on the new dataset.